MATLAB

matlab.mathworks.com

HOME PLOTS APPS FIGURE FORMAT TOOLS

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Save As | Copy Figure | Print | Show Code | FILE

Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid

LABELS AND ANNOTATIONS

/ > MATLAB Drive >

Files

Name
5G AND 6G LAB
LAB EXPERIMENT
New Folder
ASSEMENT QNO 1.m
EX : 2.m
ex 1.m
EX 2 ..m
EX 2..m
Ex 2.m
ex 5.m
ex1.m

Workspace

Name	Value	Size	Class
c	1×3001 dou...	1×3001	double
f	1×3001 dou...	1×3001	double
fc	20000	1×1	double
fm	5000	1×1	double
Fs	600000	1×1	double
idx	1426	1×1	double
m	1×3001 dou...	1×3001	double
N	3001	1×1	double
peak_freq	-1.5095e+04	1×1	double
s	1×3001 dou...	1×3001	double

EX1.m | ex2.m | ex3.m | EX4.m | untitled4.m | EX6.m

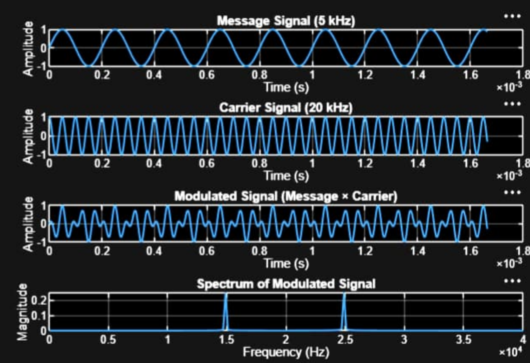
/MATLAB Drive/EX1.m

```matlab
clear; close all; clc;
%% Parameters
Fs = 600e3;          % Sampling frequency
t = 0:1/Fs:5e-3;     % 5 ms duration
fm = 5e3;            % Message frequency (≈ 5 kHz bandwidth)
fc = 20e3;           % Carrier frequency 20 kHz
%% Message signal
m = sin(2*pi*fm*t);
%% Carrier signal
c = cos(2*pi*fc*t);
%% Modulated signal (DSB-SC)
s = m .* c;
%% ---------------------- PLOTTING ----------------------

figure;

subplot(4,1,1);
plot(t(1:1000), m(1:1000), 'LineWidth',1.3);
title('Message Signal (5 kHz)');
xlabel('Time (s)'); ylabel('Amplitude');
```

Figure 1



Command Window

```
Spectrum Peak Detected at = -15.1 kHz
>>
```

36°C Partly sunny | Search | ENG IN | 14:49 03-08-2026